S P E C T R A V I D E O

U S E R S   G R O U P -- Wgtn

Newsletter Number 22 -- June 1986

The next meeting is on Monday, June 16th at 7.00 in the Staff Common Room , Wellington Clinical School , Mein Street Newtown, the next will be on 21 July 1986. (ie as usual the 3rd Monday of the month.)

Wellington SV Users Group

Committee Members

Chairman ----- Darryl Alison Ph 85522 (Paraparaumu)

Secretary ----- Joseph Dol 889275

Treasurer ----- Ross Fletcher 788454

Don Stanley 896379

Garry Clark 367872

Newsletter Committee
Steve Hanson 843495
John Matthews 327805

Other specific areas of responsibility for committee members are
Don -- Software Consultant
Garry -- Radio Access & public domain tape software;
John -- Library;
Darryl -- Program evaluation committee chairman & Kapiti users representative.

The Wellington Microcomputer Society Incorporated

Next meeting ... 10 June 1986
7:30pm at Royal Society Lecture Hall
Turnbull Street
Thorndon
Topic : Talk on Computer Security
by Bruce Goldstein from Databank

GAMES SCORES

Spectron	12910	Robert Campin
Sector Alpha	2445200	Garry Clark
Kung Fu	82435	Garry Clark
	48705	Garry Clark (*)
S C Force	748600	Deirdre Chin
Telebunny	113520	Garry Clark
	32720	Garry Clark (+)
	31670	Kalima Collinson (*)
Omac	35360	Richard Bremford
	25700	Don Stanley (+)
Ninja	251350	Nicola Tuinman
Tetra	284450	Julian Stone
Turboat	68390	Garth Thornton (*)
Turboat	104610	Taane Clark
Sasa	172950	Steve Lacey
Frantic Fred	21720	Rachel Hughes (level4)
Flipper Slip	37640	Garth Thornton
Bats Alive	3540	Colin Chin (*)
Galaxy	1447	"
Jumping Jack	2600	"
TV Nightmare	7040	"
	3640	" (*)
	4980	Deirdre Chin (+)

(+) = started in normal level
(*) - started in arcade level
otherwise practice level or unknown.

Although it is nice not to have to update the scores all the time, it would be nice to see a few more changes in the scores above. C'MON all you gamesmasters - show us how good your skills are !!

Club Sponsor

Action Computers -- P.O Box 7442, Wellington South.

Photocopying from Library

Anybody requiring articles from the library to be photocopied please send request to the LIBRARIAN, not any other committee member. We have had problems with articles not being copied when requested, please ensure that requests go to the librarian.

Also membership renewals should be sent to Box <u>26050, Newlands, Wellington</u>, not to any other box, otherwise it is possible that the mailing list will not get updated.

AND NOW FROM THE "YOU BLEW IT STEVE !!" DEPARTMENT

HEY! Whaddya know ... I've not got *anything* to put here this month. Great eh! .... So I'm of to sunny Australia, and the Islands with a clear conscience.

see ya later ... Steve

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A note to would-be contributers

Don't let *Anyone* or *Anything* put you off. The newsletter is made from your contributions, if we don't receive any, we can only produce a poor magazine.

Presently we are lacking in BASIC programs and in hints to beginners, and from beginners. We try to cater to *all* club members, but lately our we have been receiving mainly specialty items, which are of little benefit to many. Among the many things we would like to see arriving in the mail (addressed to *The Editors*) are:

- Programs - Basic, Pascal, Assembly, YouNameIt.
- Programming hints, and techniques.
- Utilities - any type, as long as they are self explanatory or come with instructions.
- News - anything you think other club members might like to know.
- Problems - we will find someone to answer them if we can't.

Everything **we receive will be considered.**

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Club Subscriptions

The club subscription is now $15.00/year

Note that expiry date for subscriptions is printed on the address label of newsletter. Subscriptions not renewed a month after this date will not receive newsletters from then.

The next couple of pages are part of a large contribution from Bob Curwood of Christchurch, which we featured in the last issue, more to come. Most of these require ASSEMBLing a Machine Code routine - using 'ZEN' or the 'Magic' assembler mentioned in previous issues. The switch routine does the same job as the Memory Expansion program - gives 328 users an extra bank without the need of a 64K card.

John

```
;********************************
;**                            **
;**     SWITCH routine V1.3    **
;**                            **
;**           by               **
;**        Bob Curwood         **
;**                            **
;**     Technical Officer      **
;**          of the            **
;** Christchurch Spectravidio  **
;**        Users Group         **
;**                            **
;**  Based on a routine in     **
;** SVI & MSX (Tasmania) by    **
;**        T Colverd           **
;********************************


;This routine enables the bank switching command
;SWITCH in SVI basic on the 328 and the 318 if
;fitted with 64k rampack.

;Assemble and run with DEFUSR=&HD23E:X=USR(0)
;The program should be protected with CLEAR [xxx],&HD23E
;Save with BSAVE"1:SWITCH",&HD23E,&HD2C6

;The original routine by T Colverd only worked under
;a cassette system. I've modified it to show which
;bank is in use each time you enter SWITCH

;When first loaded/run it copies whatever BASIC program
;you have in memory to the other bank, then goes back to
;the original.

ORG 0D23EH
LOAD 0D23EH

 REGHL: EQU 0FE5EH
 REGDE: EQU 0FE60H
HKGONE: EQU 0FF57H
 REGSP: EQU 0FE5CH
  REGA: EQU 0FE64H
  CSRY: EQU 0FA03H
CHGCLR: EQU 03750H

  JUMP: EQU 0C3H
RETURN: EQU 0C9H

LISPRT: EQU 01B05H
   CLS: EQU 03777H
  MAIN: EQU 09C4H

START:        PUSH HL
              LD A,JUMP
              LD (HKGONE),A
```

```
        LD HL,SWITCH
        LD (HKGONE+1),HL
        LD (REGSP),SP
        DI
        LD A,0FH
        OUT (88H),A
        IN A,(90H)
        LD (REGA),A
        AND 0FDH
        OUT (8CH),A
        LD HL,8000H
        LD DE,0000H
        LD BC,8000H
        LDIR
        LD A,(REGA)
        OUT (8CH),A
        EI
        POP HL
        LD A,"2"
        LD (BANK),A
        RET
SWITCH: CP RETURN
        RET NZ
        LD (REGSP),SP
        INC HL
        LD (REGHL),HL
        DI
        LD A,0FH
        OUT (88H),A
        IN A,(90H)
        LD (REGA),A
        AND 0FDH
        OUT (8CH),A
        LD HL,0000H
        LD DE,8000H
LOOP:   LD C,(HL)
        EX DE,HL
        LD B,(HL)
        LD (HL),C
        EX DE,HL
        LD (HL),B
        INC HL
        INC DE
        LD A,D
        OR E
        JR NZ,LOOP
        LD A,(REGA)
        OUT (8CH),A
        CALL INFORM
        LD SP,(REGSP)
        LD HL,MAIN
        LD A,(HL)
        POP BC
        PUSH HL
        EI
        RET
```

```
INFORM:      XOR A
             CALL CLS
             LD HL,MSG
             CALL LISPRT
              RET
MSG: DB "BANK "
BANK: DB "1",13,10,0
END
```

```
1Ø REM METRONOME
2Ø REM OLIVER  SVUG Wellington
3Ø SCREEN,Ø:WIDTH39:LOCATE,,Ø:ONSTOPGOSUB33Ø:STOPON
4Ø CLS:PRINTTAB(6)"SPECTRAVIDEO  METRONOME"
5Ø PRINTTAB(6)"~~~~~~~~~~~~~~~~~~~~~~~~~~"
6Ø PRINT:PRINT:PRINT
7Ø INPUT"Select number of BEATS to the BAR";B
8Ø PRINT:PRINT
9Ø PRINT"RIGHT cursor   for more beats per min."
1ØØ PRINT
11Ø PRINT"LEFT  cursor   for less beats per min."
12Ø PRINT:PRINT
13Ø PRINT"ESC   to  return  to  original  speed"
14Ø PRINT:PRINT
15Ø LOCATEØ,18:PRINT"METRONOME:          beats per minute"
16Ø B=B-1
17Ø E=3ØØ:GOSUB32Ø
18Ø FOR C=1 TO E:NEXT C
19Ø FOR A=1 TO B
2ØØ SOUND7,254:SOUND8,16:SOUND1,Ø:SOUNDØ,7Ø
21Ø SOUNDØ,7Ø:SOUND13,1:SOUND12,Ø:SOUND11,1ØØ
22Ø FOR C=1 TO E:NEXT C
23Ø NEXT A
24Ø SOUND8,16:SOUND1,Ø:SOUNDØ,155:SOUND13,1:SOUND12,Ø:SOUND11,1ØØ
25Ø D$=INKEY$:IFD$=""THEN18Ø
26Ø IFD$=CHR$(28)THENE=E-3:GOSUB32Ø
27Ø IFD$=CHR$(29)THENE=E+3:GOSUB32Ø
28Ø IFD$=CHR$(27)THENE=3ØØ:GOSUB32Ø
29Ø IFE<5ØTHENE=5Ø
3ØØ IFINKEY$=""THEN34Ø
31Ø GOTO26Ø
32Ø LOCATE15,18:PRINT"    ":LOCATE15,18:PRINTINT(3ØØØØ/E):RETURN
33Ø SCREENØ,1:LOCATE,,1:END
34Ø FORX=1TO5Ø:NEXT:GOTO25Ø
```

```
;*****************************
;**                         **
;**    Slow Scroll  V1.2    **
;**                         **
;**           by            **
;**       Bob Curwood       **
;**                         **
;**    Technical Officer    **
;**          of the         **
;** Christchurch Spectravidio **
;**       Users Group       **
;**                         **
;*****************************

;This program sits in the top of memory and
;when anything is printed to the crt it
;checks for the ESC key.
;If found it slows down the scrolling speed.
;The next ESC keypress toggles back to normal.

;The idea is to make it easier to read listings
;and might help with reading programs that put
;long messages to the screen.

;To use, load and run the program at the ORG.
;Then when listing etc the ESC key will toggle
;the slow scroll on/off.

        ORG OFFB6H
       LOAD OFFB6H

 CHGET: EQU 0403DH
 CHSNS: EQU 03DCAH
   ESC: EQU 0001BH

OUTDHK: EQU OFED9H

ALTRHK:         LD DE,START
                LD HL,OUTDHK+2
                LD A,D
                LD (HL),A
                LD A,E
                DEC HL
                LD (HL),A
                LD A,OC3H
                DEC HL
                LD (HL),A
START:          PUSH AF
                PUSH HL
                LD HL,(COUNT)
LOOP:           DEC HL
                LD A,H
                OR L
                JR NZ,LOOP
```

```
          CALL CHSNS
          CALL NZ,CHGET
          CP 32
          JR NZ,EXIT
          LD HL,(COUNT)
          LD A,H
          CPL
          LD H,A
          LD (COUNT),HL
EXIT:     POP HL
          POP AF
          RET
COUNT:    DW OOFEH
END
```

NOTE that this will slow down your printout/LLIST as well!

HAPPY BIRTHDAY MUSIC

```
10Ø REM HAPPY BIRTHDAY
2Ø REM Garry Clark
3Ø STOPON:ONSTOPGOSUB23Ø
4Ø COLOR 15,6,5:SCREENØ,Ø
5Ø INPUT"Whose BIRTHDAY is it ";N$
6Ø L=LEN(N$)
7Ø SCREEN1:REM WORDING
8Ø L=LEN(N$):C$=""
9Ø FORZ=1TOL:B$=MID$(N$,Z,1)
10Ø C$=C$+B$+" ":NEXT
11Ø FORY=1TO2:FORX=1TO2
12Ø LOCATE45+X,8Ø+Y
13Ø PRINT"H A P P Y   B I R T H D A Y"
14Ø LOCATE128+X-L*6,11Ø:PRINTC$
15Ø NEXTX,Y
16Ø REM MUSIC Happy Birthday
17Ø SOUND7,22Ø:SOUND6,3:'Channel 3 Drums
18Ø PLAY"T255","T255","T255"
19Ø PLAY"M35ØØS1L8FFL4GFB-L2A L8FFL4GF05CL204B- L8FFL405FD04
B-AG L805E-E-L4D04B-05C04B-R"
2ØØ PLAY"M5ØØØ05S1L8FFL4GFB-L2A L8FFL4GF06CL205B- L8FFL406FD
05B-AG L806E-E-L4D05B-06C05B-R","M25ØØ06S3L8RRL4RRB-AR L8RRL
4RF07CL206R L8RRL406RR B-AR L807E-E-L4R"
21Ø PLAY"M4ØØØS1L8FFL4GFB-L2A L8FFL4GF05CL204B- L8FFL405FD04
B-AG L805E-E-L4D04B-05C04B-R","M4ØØØS1L803FFL4GFB-L2AL8FFL4G
F03CL203B-L8FFL404FD02B-AGL803E-E-L4D06B-07C07L2B-","M25ØØ05
S1L8RRL4RRBBR L8RRL4RBBRR L8RRL4RRBBRL8BBL4R"
22Ø IFINKEY$=""THEN22ØELSE16Ø
23Ø COLOR 15,4,5:SCREENØ,1:END
```

CP/M-BASIC File Transfers Revisited

John Bridson

The November 1984 issue of the Club newsletter contained an article in which Peter Thakurdas described a method he had developed for transferring disk files in both directions between CP/M and BASIC. The BASIC programs involved, BASIC.IT and CPM.IT, were duly stored on one of my BASIC disks, without checking them for typing errors.

It was only very recently that I came back to these programs to get them working. Having done so, I decided that they could do with having their operating instructions built in as screen messages, instead of having to refer to written details elsewhere. I also wanted to be able to use either disk drive as the source or destination of the file being transferred.

With a basic idea of how the programs worked, I began adding to them, testing each as I went. All went well at first but then BASIC began throwing up odd "Undefined line number" errors when the line was really there! With no other obvious reason for the problems, I began to suspect that the machine code was overwriting the BASIC program in memory.

Using information in the early chapters of "The Magic of Spectravideo" to check BASIC's memory map addresses, I confirmed that the machine code in my last "working" version was indeed being written to memory perilously close to the end of the program itself. A safer place for the machine code was obviously needed and the best location was above the memory area reserved for BASIC. However, system data stored in this area had to be protected in its turn and this meant that the default Top of Memory setting had to be lowered slightly to provide a clear area for the machine code. The CLEAR statement in BASIC can be used to do just this.

As originally described, the CP/M file G8000.COM was required during the transfer from CP/M to BASIC. G8000.COM is a modified version of REBOOT.COM and it is still needed. If you don't have it yet, you can create it quite easily, by using DDT on REBOOT.COM. Follow the procedure below

(1) at the CP/M prompt, type

```
DDT REBOOT.COM
```

(2) at the DDT prompt, type

```
S10D
00
.
G0        (or CTRL-C)
```

(3) at the CP/M prompt, type

```
SAVE 1 G8000.COM
```

and it's done!

The modified BASIC programs were developed on a MkI SV328 system, for use with an 80 column display. I tested them by moving a 24917 byte file from

BASIC to CP/M and a 16336 byte file back from CP/M to BASIC. They each took some time to complete but they were both successful and, apparently, without any errors. The new versions of both programs are listed below.

BASIC.IT

```
1 '-- BASIC.IT -- Public Domain Software for SV 328 Disk System
2 ' by Peter Thakurdas, 1/11/84 (modified by John Bridson, 14-5-86)
100 CLS:PRINT ,,SPC(5) "┌──────────┐":PRINT ,,SPC(5) "│  BASIC.IT  │":PRI
NT ,,SPC(5) "└──────────┘":LOCATE 0,5
110 PRINT "This program transfers 16,384 (16K) bytes of data from CP/M memo
ry to BASIC's
120 PRINT "RAM area, ready for transfer to a BASIC disk.":PRINT
130 PRINT "Start the process in CP/M, by running G8000.  Then use DDT <[b:]
filename> to
140 PRINT "load the data file into memory.  When this has been done, note t
he hex number
150 PRINT "under 'NEXT'.  Replace the CP/M disk in drive A with a BASIC Sys
tem disk.
160 PRINT "Type G8000 at DDT's hyphen prompt and wait while BASIC boots up.
 Then RUN this
170 PRINT "program and follow the prompts given.
180 PRINT:PRINT:PRINT "Press <ESC> to quit, any other key to continue...";
190 K$=INPUT$(1):IF K$=CHR$(27) THEN END
200 CLEAR 200,54665!:GOSUB 700:' Initialise machine code routine
210 X=USR(0):' Transfer 16K data block from 0100H to 9000H
220 CLS:PRINT "A 16K block of data has been moved from CP/M memory (0100H t
o 40FFH) to BASIC's
230 PRINT "RAM area, starting at 9000H.
240 PRINT:INPUT "Enter hex number under 'NEXT' when DDT loaded the data";N$
250 B=&H9000:E=B+VAL("&H"+N$)-&H101
260 PRINT:INPUT "Enter filename to use for transferred data";F$
270 PRINT:PRINT "Which drive is '";F$;"' to be stored on (1 or 2)? ";
280 D$=INPUT$(1):IF D$<>"1" AND D$<>"2" THEN 280 ELSE PRINT D$:D$=D$+":"
290 PRINT:PRINT "Save data in text form (y/n)? ";
300 K$=INPUT$(1):K$=CHR$(ASC(K$) AND 95):IF K$<>"N" AND K$<>"Y" THEN 300
310 PRINT K$:PRINT:IF K$="Y" THEN 340
320 ' Write machine code to disk
330 PRINT "Using BSAVE to store data... ";:BSAVE D$+F$,B,E:PRINT:GOTO 390
340 ' Write text data to disk
350 OPEN D$+F$ FOR OUTPUT AS #1
360 FOR I=B TO E
370  A$=CHR$(PEEK(I)):PRINT A$;:PRINT #1,A$;:IF A$=CHR$(26) THEN E=I:GOTO 3
90
380 NEXT I
390 CLOSE:PRINT:PRINT E+1-B;"bytes written to disk":PRINT:CLEAR 200,54712!:
END
700 ' Routine moves data from low CP/M RAM area to BASIC's RAM
710 J=&HD590:' Code location
720 READ A$:IF A$="END" THEN 820 ELSE POKE J,VAL("&H"+A$):J=J+1:GOTO 720
730 ' Machine code (HEX) follows
740 DATA F3,3E,0F,D3,88,DB,90,E6,FD,D3,8C:' ROM disable
750 DATA 11,00,90:' LD DE,9000H - destination addr
760 DATA 21,00,01:' LD HL,0100H - source addr
770 DATA 01,00,40:' LD BC,4000H - byte count
780 DATA ED,B0    :' LDIR - transfer data
```

```
790 DATA 3E,0F,D3,88,DB,90,F6,02,D3,8C,FB:' ROM enable
800 DATA C9 :' return to BASIC
810 DATA END
820 DEF USR=&HD590:RETURN
```

CPM.IT

```
1 '-- CPM.IT -- Public Domain Software for SV328 Disk System
2 ' by Peter Thakurdas, 1/11/84 (modified by John Bridson, 14-5-86)
100 CLS:PRINT ,,SPC(6) "┌──────────┐":PRINT ,,SPC(6) "│  CPM.IT  │":PRINT ,
,SPC(6) "└──────────┘":LOCATE 0,5
110 PRINT "This program transfers an ASCII file from a BASIC disk to the CP
/M RAM area,
120 PRINT "ready to be SAVEd to a CP/M disk.":PRINT
130 PRINT "The file to be transferred can be no longer than 32,500 bytes (j
ust less than
140 PRINT "8 disk blocks in BASIC).  If it is a BASIC program file, it shou
ld have been
150 PRINT "stored in ASCII form, using SAVE ";CHR$(34);"d:filename";CHR$(34
);",A."
160 PRINT:PRINT:PRINT "Press <ESC> to quit, any other key to continue... ";
170 K$=INPUT$(1):IF K$=CHR$(27) THEN END ELSE LOCATE 0,13:PRINT SPC(50)
180 CLEAR 200,54665!:GOSUB 600:' Initialise machine code
190 ' Transfer data to low CP/M RAM
200 LOCATE 0,13:INPUT "Which file is to be transferred";F$
210 PRINT:PRINT "Which drive is '";F$;"' on (1 or 2)? ";
220 D$=INPUT$(1):IF D$<>"1" AND D$<>"2" THEN 220 ELSE PRINT D$:D$=D$+":"
230 PRINT:PRINT:N=0:OPEN D$+F$ FOR INPUT AS #1
240 N=N+1:IF NOT EOF(1) THEN A$=INPUT$(1,#1):PRINT A$;:GOSUB 500:GOTO 240
250 CLOSE:A$=CHR$(26):GOSUB 500:PRINT:PRINT N;"bytes transferred to CP/M RA
M area
260 PRINT:PRINT:PRINT "Now replace the BASIC disk in drive A with a CP/M Sy
stem disk.  When CP/M has
270 PRINT "been booted up, type
280 PRINT:PRINT SPC(6) "save";INT(N/256)+1;"<[b:]filename>  <ENTER>"
290 PRINT:PRINT "Press any key when ready to continue... ";
300 A$=INPUT$(1):DEF USR=0:PRINT USR(0):END
500 POKE &HD58D,ASC(A$):X=USR(0):RETURN:' Write char to RAM
600 ' Routine moves data from BASIC to low CP/M RAM area
610 J=&HD590:' Machine code location
620 READ A$:IF A$="END" THEN 730 ELSE POKE J,VAL("&H"+A$):J=J+1:GOTO 620
630 ' Machine code (HEX) follows
640 DATA F3,3E,0F,D3,88,DB,90,E6,FD,D3,8C:' ROM disable
650 DATA ED,5B,8E,D5:' LD DE,(D58Eh) - destination addr
660 DATA 3A,8D,D5    :' LD A,(D58Dh)  - get data
670 DATA 12          :' LD (DE),A     - transfer data
680 DATA 13          :' INC DE        - update destination addr
690 DATA ED,53,8E,D5:' LD (D58Eh),DE - save new destination addr
700 DATA 3E,0F,D3,88,DB,90,F6,02,D3,8C,FB:' ROM enable
710 DATA C9 :' return to BASIC
720 DATA END
730 POKE &HD58E,0:POKE &HD58F,1:' initial CP/M address is 0100H
740 DEF USR=&HD590:RETURN
```

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SPECTRAVIDEO USERS GROUP (Wgtn)

Membership Form

Name ____

Address ____

____

____

Phone (home) ___________ (Work) __________________

Interests (Please tick relevant ones)

Basic __ Assembler __ Games __ CP/M __ Education __

Work Use __ Other (please specify) __

Meetings (please tick relevant one)

Usually __ Sometimes __ Rarely __ Newsletter Only __

O F F I C E U S E O N L Y

Paid	Date	Expires	Signed	Number

Send Completed Form (Whole Page) plus #15.00 to

SpectraVideo Users Group (WGTN)

P.O Box 26050 , Newlands , Wellington

WGTN SV USERS GROUP

Bug Report Form

Please fill in the following and send to BUGS, P.O Box 26050, Newlands, Wellington

Only 1 bug per form please

---

1) Where bug occurs

a) Basic Command --> Which Command ______________________

b) Commercial Basic Program --> Program Name ___________

Line Number if applicable __________

c) Commercial Machine Code Program --> Program Name ________

d) CP/M Program --> Program Name ____________________

e) Hardware --> Hardware Name and Number ______________

f) Manual / Book --> Title ___________________________

Author ____________________________ Page ________

g) Other .. Please specify

****

2) Describe the bug , and under what circumstances it occurs if possible

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++

3) If You have a fix for the bug please decribe it here or on a separate sheet

Public Domain CP/M

Anyone wanting anything from the library of CP/M PD software, please write to the Librarian at the club address, specifying either which disc, or which programs you would like. As my drives are single sided, a whole disc from the list now occupies up to 2 full discs. Most of the programs on the discs are *'Squeezed'*, this means they have been run through a function in New Sweep (NSW, Nsweep etc. but not the earlier Sweep) so as to take up as little room as possible. They all have a 'Q' as the 2nd letter in their extension, and can be easily unsqueezed by using New Sweep to tag them and then the 'Q' option to unsqueeze them, usually onto another disc.

I have had some enquiries for a description of the programs, rather than just their title's, and have now seen them myself, here goes:

Disc 1:

AD.COM - The original 'Adventure' game. Needs the following files on the disc (unsqueezed) to run: AD.COM, ADVENTUR.MSG, COMMON.DAT, ATAB.DAT, KTAB,DAT, LTEXT.DAT, RTEXT.DAT, STEXT.DAT, and TRAVEL.DAT. It works, but is slow and accesses the disc too much.

CALENDAR.COM - Prepares the year's calendar to a disc file, from which it can be printed.

SVMBASIC.COM - the Microsoft Basic the northern hemisphere apparently received on their distribution disc. Has features like the ability to load and run files while leaving variables intact, but has no graphics, and no full-screen editting.

TTT.ASM & .COM - Tic-Tac-Toe. The COM file won't work, as it is set up for Osborne screen commands. The ASM file can be adapted easily though and re-assembled.

Disc 2:

BC.ASM - a 'Binary Calculator', works in Hex, Decimal, and Octal too.

DBL.ASM + others - Reads an ASCII file and prints it in two columns, using compressed print. Can be set up and re-assembled for practically any printer with compressed print.

LU3.COM - Library utility, for putting lots of files into one large file, saving wasted disc and directory space.

MX*.* - Utilities to set up Epson MX printers.

PATCHER.COM, WS.PATCH - Forget it unless you have an Osborne and want a blinking cursor in Wordstar etc.

PSWORD.ASM & COM - Puts a password routine into a program, if the right password is entered, drops program to right location and runs it.

VDO.COM - Text editor. Full instructions included.

WSVIEW.C & COM - written in *C*, views Wordstar files without garbage. Later versions (2.04+) of New Sweep do this anyway.

XMOD90.ASM - Modem program for mainframe terminals.

Disc 3: dBaseII files.

Most of these are part of a rather complete Geneology package. But others are for other databases, and some are dBaseII utilities. KITCH*.HBL + CODES.HBL are parts of a recipe data base, while PHONE*.CMD is a phone number database, and START.CMD is the starting menu for both.

Disc 4: Microsoft Basic files.

These are basic programs, some out of magazines (BYTE) and some by other users. Some are games, some mathematical or statistical, some utilities. Many of them are very good.

Disc 5: CP/M Utilities

Most of these are assembly language programs, some of them won't work on our computer, others need only a few changes, most work 'as is'.

AUTO*.* - Auto run program, doesn't seem to work.
BANNER.ASM - Prints a banner from a string, either to printer or to *Paper Tape*.
FBAD56.COM - finds and blocks off bad sectors on a disc.
LOCATE.ASM - relocates programs in memory.
MEX.COM - Modem Executive. Set up for the Spectravideo, has an extensive inbuilt help routine, which uses MEX.HLP.
SAP36.ASM & COM - rewrites directory to disc in alphabetical order. Deleted filenames are blanked over to make disc editting (with DU - see later) easier.
CLOSE.ASM - rewrites directory in lower case so that CP/M can't access the files from the command level.
OPEN.ASM - undoes the above, if you know the password included during assembly. The COM file of this is not locked off by CLOSE.
SCH*.* - American Income Tax worksheets from SuperCalc.
SCRAMBLE.ASM - XOR's a file with an eight letter password, the file is then useless until the program is run again with the same password.
SPACEIN.ASM - Space Invaders, without graphics.
UNERA15.ASM & COM - recovers deleted files, as long as they haven't been overwritten since they were deleted.

Disc 6:

ALFAROM - Alphabetical listing of the ROM entry points, compiled by Deane Whitmore. NUMROM the above in numeric order.
*.HQP - Files used by HELP.COM, they give a short help on the topic of their title. You can create your own to use HELP.
DU-V83.COM - Disc Utility. This is a disc editor. It reads and writes sectors, directories and much more.
FIND40.COM - Finds strings in files.
IF/SKIP.COM - Submit utilities to enable branching.
RECLAIM.COM - Like UNERA but menu driven, and complex to use.
SUPERSUB.COM - An enhanced version of SUBMIT.
Z80DET - Details of the Z80 Assembly code instruction set.

Disc 7:

ADVENTUR - Similar to the one on Disc 1, but this only needs two files to run: ADVENTUR.COM and ADVENTUR.WRK. According to the documentation file, there is supposed to be a .HEX file which changes the .DAT file (which can be editted to change the messages) into a .WRK file, but we don't have it.

AIM.COM - Automatic Investment Management. Keep track of shares etc. Gives advise on when to buy/sell.

CALENDAR.CAL - SuperCalc2 worksheet, uses the Osborne inbuilt clock to provide a calendar. May be convertible if the time and date are input manually.

PC*.COM - A Database creation/management system for the SV. Menu driven and designed to be easy to use.

PI.COM - Uses the binomial theorem to find the upper and lower bounds on the value of PI.

Disc 8: Mumps

Anyone out there who knows how to program in MUMPS is welcome to it, theres nothing here to explain it and it's hard to get to work.

Bug Corrections

Further to the correction on page 10 of the April issue. The solution shown will solve the problem, but there is a better way. When BASIC boots up it puts aside a default 200 Bytes of string space. If you create a string, or a number of strings totalling greater than 200 bytes you get an 'Out of string space' error, unless you have executed a *CLEAR n* command, the *n* standing for the number of bytes BASIC is to set aside for string storage. A value of 500 will be enough to make the program run as originally written - add *5 CLEAR 500* to the start of the program.

The code on page 10 for a Wordstar busy routine should start with E5.

John.

Graphics from CP/M

Don's program in the JAN/FEB 86 newsletter can be written another way, as I have learned from an article by S.W. McNamee in the South Australian newsletter.

Resident in CP/M is a routine which does exactly what Don's routine does, including switching off all the interrupts. CP/M uses this to access the ROM routines for console input etc. All you need to do is load the address of the location in ROM you want to call, and then call E651. You still have to use your own routine to set up the graphics screen, but then, after changing location FE3A to 1 (let basic's routines know it is in graphics mode) call the CLS routine at 459A by an inline statement:

```
    inline($21/$9A/$45  {LD HL,459A}
           $cd/$51/$e6  {CALL e651}    );
```

The same thing can be used for the LINE and PAINT routines. Line would become :

```
procedure Line(left_x,top_y,right_x,bottom_y,colour:byte;
                                              Style: char);
var location:integer;
begin
  case style of
   'B': location:=$2452;
   'L': location:=$243c;
   'F': location:=$2401;
   else exit; {exit Line if invalid style}
  if not (colour in [0..15]) then exit; {invalid colour}
  mem[$fe3a]:=1;  {Screen mode = graphics}
  mem[$fa13]:=colour;
  mem[$fe43]:=0;
  mem[$fe47]:=0;
  mem[$fe42]:=right_x;
  mem[$fe46]:=right_x;
  mem[$fe45]:=0;
  mem[$fe49]:=0;
  mem[$fe44]:=bottom_y;
  mem[$fe48]:=bottom_y;
  mem[$fe38]:=$c9;
  inline($06/$00/     {LD B,0}
         $3a/left_x/  {LD A,left_x}
         $4f/         {LD C,A}
         $16/$00/     {LD D,0}
         $3a/top_y/   {LD A,top_y}
         $5f/         {LD E,A}
         $3e/$00/     {LD A,0}
         $32/$2b/$fe/{LD (fe2b),A}
         $2a/location{LD HL,location}
         $CD/$51/$e6 {CALL e651}    );
  mem[$fe38]:=$c3;
end;
```

This could also be achieved from SVMBASIC with pokes and DEF USR calls, possibly in strings as in issue 9 page 14.

Also in the article were some other interesting facts about our computer, among them:

The 80 column card is accessed directly as memory after being bank-switched in. First you disable the interrupts (DI) then you send 255 out port 58 hex. The 80 Column card can then be addressed at F000 to F77F, you can both read and write to it. It is switched out by sending 0 to port 58h, the interrupts can then be re-enabled. This can only be done in machine code not using the ROMs as this is the same area as Basic has its work space - ie stacks etc. Also beware any routines which also disable/reenable the interrupts. In Turbo Pascal you could write a program like:

```
program copy_screen;
var screen:absolute $f000;
    backup:array[$f000..$f77f] of byte;
    key:char;
begin
  inline($f3); {di}
  port[$58]:=$ff;
  move(screen,backup,$780);
  port[$58]:=0;
  inline($fb); {EI}
  clrscr;
  write('Press any key :');
  repeat until keypressed;
  read(kbd,key);
  write(key);
  inline($f3);
  port[$58]:=$ff;
  move(backup,screen,$780);
  port[$58]:=0;
  inline($fb);
end.
```
